35.5: Growth, morphogenesis, and cell differentiation produce the plant body

The specific series of changes by which cells form tissues, organs, and organisms is called **development**. Development unfolds according to the genetic information that an organism inherits from its parents but is also influenced by the external environment. A single genotype can produce different phenotypes in different environments. For example, the aquatic plant *Cabomba aquatica* forms two very different types of leaves, depending on whether the shoot apical meristem is submerged **(Figure 35.25)**. This ability to alter form in response to local environmental conditions is called *developmental plasticity*. Dramatic examples of plasticity, as in *Cabomba*, are much more common in plants than in animals and may help compensate for plants' inability to escape adverse conditions by moving.

Figure 35.25

Developmental plasticity in the aquatic plant *Cabomba aquatica*.

Cabomba aquatica is a plant in water with a surface leaf and underwater leaves.

The underwater leaves of Cabomba are feathery, an adaptation that protects them from damage by lessening their resistance to moving water. In contrast, the surface leaves are pads that aid in flotation. The two leaf types have genetically identical cells, but their different environments result in the turning on or off of different genes during leaf development.

The three overlapping processes involved in the development of a multicellular organism are growth, morphogenesis, and cell differentiation. *Growth* is an irreversible increase in size. *Morphogenesis* (from the Greek *morphê*, shape, and *genesis*, creation) is the process that gives a tissue, organ, or organism its shape and determines the positions of cell types. *Cell differentiation* is the process by which cells with the same genes become different from one another. We'll examine these three processes in turn, but first we'll discuss how applying techniques of modern molecular biology to model organisms, particularly *Arabidopsis thaliana*, has revolutionized the study of plant development.

Model Organisms: Revolutionizing the Study of Plants

As in other branches of biology, techniques of molecular biology and a focus on model organisms such as *Arabidopsis thaliana* have catalyzed a research explosion in the last few decades. *Arabidopsis*, a tiny weed in the mustard family, has no inherent agricultural value but is a favored model organism of plant geneticists and molecular biologists for many reasons. It is so small that thousands of plants can be cultivated in a few

square meters of lab space. It also has a short generation time, taking about six weeks for a seed to grow into a mature plant that produces more seeds. This rapid maturation enables biologists to conduct genetic cross experiments in a relatively short time. One plant can produce over 5,000 seeds, another property that makes *Arabidopsis* useful for genetic analysis.

Beyond these basic traits, the plant's genome makes it particularly well suited for analysis by molecular genetic methods. The *Arabidopsis* genome, which includes about 27,000 protein-encoding genes, is among the smallest known in plants. Furthermore, the plant has only five pairs of chromosomes, making it easier for geneticists to locate specific genes. Because *Arabidopsis* has such a small genome, it was the first plant to have its entire genome sequenced.

The natural range of *Arabidopsis* includes varied climates and elevations, from the high mountains of Central Asia to the European Atlantic coast, and from North Africa to the Arctic Circle. These local varieties can differ markedly in outward appearance **(Figure 35.26)**. Genome-sequencing efforts are being expanded to include hundreds of populations of Arabidopsis from throughout its natural range in Eurasia. Contained in the genomes of these populations is information about evolutionary adaptations that enabled Arabidopsis to expand its range into new environments following the retreat of the last ice age. This information may provide plant breeders with new insights and strategies for crop improvement.

Figure 35.26

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    Variations in leaf arrangement, leaf shape, and shoot growth
between different populations of *Arabidopsis thaliana*.
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Photo showing the variability in leaf arrangements, shoot growth, and leaf shapes in Arabidopsis thaliana.

Information in the genomes of these populations may provide insights into strategies for expanding crop production into new environments.

Another property that makes *Arabidopsis* attractive to molecular biologists is that its cells can be easily transformed with *transgenes*, genes from a different organism that are stably introduced into the genome of another. CRISPR technology (see Figure 20.14), which is rapidly becoming the technique of choice for creating plants with specific mutations, has been used successfully in *Arabidopsis*. By disrupting or "knocking out" a specific gene, scientists can garner important information about the gene's normal function.

Large-scale projects are under way to determine the function of every gene in *Arabidopsis*. By identifying each gene's function and tracking every biochemical pathway, researchers aim to determine the blueprints

for plant development, a major goal of systems biology. It may one day be possible to make a computer-generated “virtual plant” that enables researchers to visualize which genes are activated in different parts of the plant as the plant develops.

Basic research on model organisms such as *Arabidopsis* has accelerated the pace of discovery in the plant sciences, including the identification of the complex genetic pathways underlying plant structure. As you read more about this, you’ll be able to appreciate not only the power of studying model organisms but also the history of investigation that underpins all modern plant research.

Interview

Interview with Joanne Chory: Sequencing the *Arabidopsis* **genome****

A photo of Joanne Chory.

Growth: Cell Division and Cell Expansion

Cell division enhances the potential for growth by increasing the number of cells, but plant growth itself is brought about by cell enlargement. The process of plant cell division is described more fully in Chapter 12 (see Figure 12.10), and Chapter 39 discusses the process of cell elongation (see Figure 39.7). Here we are concerned with how cell division and enlargement contribute to plant form.

Cell Division

The new cell walls that bisect plant cells during cytokinesis develop from the cell plate (see Figure 12.10). The precise plane of cell division, determined during late interphase, usually corresponds to the shortest path that halves the cytoplasm of the parent cell. However, during certain points in development the cytoplasm may not be divided equally, resulting in one daughter cell being larger than the other, even though they have the same number of chromosomes. Such cases of *asymmetrical cell division* usually signal a key event in development. For example, the formation of guard cells involves an asymmetrical cell division. An epidermal cell divides asymmetrically, forming a large cell that remains an undifferentiated epidermal cell and a small cell that becomes the guard cell “mother cell.” Guard cells form when this small mother cell divides in a plane perpendicular to the first cell division **(Figure 35.27)**. Thus, asymmetrical cell division generates cells with different fates—that is, cells that mature into different types.

Figure 35.27

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Asymmetrical cell division and stomatal development.
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Asymmetrical cell division is shown with an unspecialized epidermal cell dividing to form a guard cell mother cell. The mother cell then divides perpendicular to the first cell division.

An asymmetrical cell division precedes the development of epidermal guard cells, the cells that border stomata (see Figure 35.18).

Cell Expansion

Cell divisions do not constitute growth because there is no increase of mass involved. Rather, it is cell expansion that is responsible for plant growth. Before discussing how cell expansion contributes to plant growth and form, it is useful to consider the difference in cell expansion between plants and animals. Animal cells grow mainly by synthesizing protein-rich cytoplasm, a metabolically expensive process. Growing plant cells also produce additional protein-rich material in their cytoplasm, but water uptake typically accounts for about 90% of expansion. Most of this water is stored in the large central vacuole. The vacuolar solution, or *vacuolar sap*, is very dilute and nearly devoid of the energetically expensive macromolecules that are found in great abundance in the rest of the cytoplasm. Large vacuoles are therefore a “cheap” way of filling space, enabling a plant to grow rapidly and economically. Bamboo shoots, for instance, can elongate more than 2 m per week. Rapid and efficient extensibility of shoots and roots was an important evolutionary adaptation that increased their exposure to light and soil.

Plant cells rarely expand equally in all directions. Their greatest expansion is usually oriented along the plant’s main axis. For example, cells near the tip of the root may elongate 20 times or more their original length, with relatively little increase in width. The orientation of cellulose microfibrils in the innermost layers of the cell wall causes this differential growth. The microfibrils do not stretch, so the cell expands mainly perpendicular to the main orientation of the microfibrils, as shown in **Figure 35.28**. A leading hypothesis proposes that microtubules positioned just beneath the plasma membrane organize the cellulose-synthesizing enzyme complexes and guide their movement through the plasma membrane as they create the microfibrils that form much of the cell wall.

Figure 35.28

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The orientation of plant cell expansion.
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Cellulose microfibrils are oriented left to right while cross-links in the cell are oriented up and down. Expansion happens vertically here. The scale is 5 micrometers.

Growing plant cells expand mainly through water uptake. In a growing cell, enzymes weaken cross-links in the cell wall, allowing it to expand as water diffuses into the vacuole by osmosis; at the same time, more microfibrils are made. The orientation of the cell expansion is mainly perpendicular to the orientation of cellulose microfibrils in the wall. The orientation of microtubules in the cell's outermost cytoplasm determines the orientation of cellulose microfibrils (fluorescent LM). The microfibrils are embedded in a matrix of other (noncellulose) polysaccharides, some of which form the cross-links visible in the TEM.

Morphogenesis and Pattern Formation

A plant's body is more than a collection of dividing and expanding cells. During morphogenesis, cells acquire different identities in an ordered spatial arrangement. For example, dermal tissue forms on the exterior and vascular tissue in the interior—never the other way around. The development of specific structures in specific locations is called **pattern formation**.

Two types of hypotheses have been put forward to explain how the fate of plant cells is determined during pattern formation. Hypotheses based on *lineage-based mechanisms* propose that cell fate is determined early in development and that cells pass on this destiny to their progeny. In this view, the basic pattern of cell differentiation is mapped out according to the directions in which meristematic cells divide and expand. On the other hand, hypotheses based on *position-based mechanisms* propose that the cell's final position in an emerging organ determines what kind of cell it will become. In support of this view, experiments in which neighboring cells have been destroyed with lasers have demonstrated that a plant cell's fate is established late in the cell's development and largely depends on signaling from its neighbors.

In contrast, cell fate in animals is largely determined by lineage-dependent mechanisms involving transcription factors. The homeotic (*Hox*) genes that encode such transcription factors are critical for the proper number and placement of embryonic structures, such as legs and antennae, in the fruit fly *Drosophila* (see Figure 18.19). Interestingly, maize has a homolog of *Hox* genes called KNOTTED-1, but unlike its counterparts in the animal world, KNOTTED-1 does not affect the number or placement of plant organs. As you will see, an unrelated class of transcription factors called *MADS-box* proteins plays that role in plants. KNOTTED-1 is, however, important in the development of leaf shape, including the production of compound leaves. If the KNOTTED-1 gene is expressed in greater quantity than normal in the genome of tomato plants, the normally compound leaves will then become "super-compound" **(Figure 35.29)**.

Figure 35.29

Overexpression of a *Hox*-like gene in leaf formation.

The left leaf has a few leaves extending from either side of its main stem. The right leaf has leaves extending from leaves extending from leaves, and so on.

KNOTTED-1 is a gene that is involved in leaf and leaflet formation. An increase in its expression in tomato plants results in leaves that are "super-compound" (right) compared with normal leaves (left).

Gene Expression and the Control of Cell Differentiation

The cells of a developing organism can synthesize different proteins and diverge in structure and function even though they share a common genome. If a mature cell removed from a root or leaf can dedifferentiate in tissue culture and give rise to the diverse cell types of a plant, then it must possess all the genes necessary to make any kind of cell in the plant. Therefore, cell differentiation depends, to a large degree, on the control of gene expression—the regulation of transcription and translation, resulting in the production of specific proteins.

Evidence suggests that the activation or inactivation of specific genes involved in cell differentiation results largely from cell-to-cell communication. Cells receive information about how they should specialize from neighboring cells. For example, two cell types arise in the root epidermis of *Arabidopsis*: root hair cells and hairless epidermal cells. Cell fate is associated with the position of the epidermal cells relative to other plant cells. The immature epidermal cells that are in contact with two underlying cells of the root cortex differentiate into root hair cells, whereas the immature epidermal cells in contact with only one cell in the cortex differentiate into mature hairless cells. The differential expression of a homeotic gene called GLABRA-2 (from the Latin *glaber*, bald) is needed for proper distribution of root hairs **(Figure 35.30)**. Researchers have demonstrated this requirement by coupling the GLABRA-2 gene to a "reporter gene" that causes every cell expressing GLABRA-2 in the root to turn pale blue following a certain protocol. The GLABRA-2 gene is normally expressed only in epidermal cells that will not develop root hairs.

Figure 35.30

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    Control of root hair differentiation by a homeotic gene.
(LM)
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Control of root hair differentiation by GLABRA-2, showing root cap cells, epidermal cells and cortical cells. Scale of image: 20 micrometers.

The image shows cross section of root with different cell types. Root cap cells are smallish cells in a single layer on the outer edge of root, external to the epidermal layer, and will be sloughed off before root hairs emerge. Inside the root cap cells are larger epidermal and cortical cells. When an epidermal cell borders two cortical cells, homeotic gene GLABRA-2 is not expressed, and the cell with develop a root hair. When an epidermal cell borders a single cortical cell, the homeotic gene GLABRA-2 is expressed, and the cell remains

hairless. Expression of GLABRA-2 can be determined in this image by the presence of a stain in cells where the gene is expressed.

WHAT IF? *What would the roots look like if* GLABRA-2 *were rendered dysfunctional by a mutation?*

*For suggested answer, see*Appendix A.*

Interview

Interview with Philip Benfey: Studying root development

A photo of Philip Benfey.

Shifts in Development: Phase Changes

Multicellular organisms generally pass through developmental stages. In humans, these are infancy, childhood, adolescence, and adulthood, with puberty as the dividing line between the nonreproductive and reproductive stages. Plants also pass through stages, developing from a juvenile stage to an adult vegetative stage to an adult reproductive stage. In animals, the developmental changes take place throughout the entire organism, such as when a larva develops into an adult animal. In contrast, plant developmental stages, called *phases*, occur within a single region, the shoot apical meristem. The morphological changes that arise from these transitions in shoot apical meristem activity are called **phase changes**. In the transition from a juvenile phase to an adult phase, some species exhibit conspicuous changes in leaf shape **(Figure 35.31)**. Juvenile nodes and internodes retain their juvenile status even after the shoot apical meristem of the main shoot has changed to the adult phase. Therefore, any *new* leaves that develop on branches that emerge from axillary buds at juvenile nodes will also be juvenile, even though the apical meristem of the stem's main axis may have been producing mature nodes for years.

Figure 35.31

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Phase change in *Muehlenbeckia australis*.
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A juvenile leaf is fiddle or frog-shaped with a shiny surface while a mature leaf is oval and smooth.

This native climbing vine of New Zealand has two types of leaves: fiddle-shaped juvenile leaves (top) and oval, smooth-margined mature leaves. This dual foliage reflects a phase change in the development of the

apical meristem of each shoot. Once a node forms, the developmental phase—juvenile or adult—is fixed; fiddle-shaped leaves do not mature into oval-shaped leaves.

If environmental conditions permit, an adult plant is induced to flower. Biologists have made great progress in explaining the genetic control of floral development—the topic of the next section.

Genetic Control of Flowering

Flower formation involves a phase change from vegetative growth to reproductive growth. This transition is triggered by a combination of environmental cues, such as day length, and internal signals, such as hormones. (You will learn more about the roles of these signals in flowering in Concept 39.3.) Unlike vegetative growth, which is indeterminate, floral growth is usually determinate: The production of a flower by a shoot apical meristem generally stops the primary growth of that shoot. The transition from vegetative growth to flowering is associated with the switching on of flower-inducing genes. The protein products of these genes are transcription factors that regulate the genes required for the conversion of the indeterminate vegetative meristems to determinate floral meristems.

When a shoot apical meristem is induced to flower, the order of each primordium's emergence determines its development into a specific type of floral organ—a sepal, petal, stamen, or carpel (see Figure 30.8 to review basic flower structure). These floral organs form four whorls that can be described roughly as concentric "circles" when viewed from above. Sepals form the first (outermost) whorl; petals form the second; stamens form the third; and carpels form the fourth (innermost) whorl. Plant biologists have identified several genes that encode transcription factors that regulate the development of this characteristic floral pattern. Positional information determines which genes are expressed in a particular floral organ primordium. The result is the development of an emerging floral primordium into a specific floral organ. A mutation in a flower-inducing gene can cause abnormal floral development, such as petals growing in place of stamens **(Figure 35.32)**. Some homeotic mutants with increased petal numbers produce showier flowers that are prized by gardeners.

Figure 35.32

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Genes and pattern formation in flower development.
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Two photos compare a normal Arabidopsis flower with an abnormal flower.

Normal flower: Arabidopsis normally has four whorls of flower parts: sepals, petals, stamens, and carpels.

Abnormal flower: Researchers have identified several mutations of organ identity genes that cause abnormal flowers to develop. This Arabidopsis flower has an extra set of petals in place of stamens and an internal flower where normal plants have carpels.

MAKE CONNECTIONS *Provide another example of a homeotic gene mutation that leads to organs being produced in the wrong place (see*Concept 18.4*).**

*For suggested answer, see**Appendix A.***

By studying mutants with abnormal flowers, researchers have identified and cloned three classes of genes that determine floral organ identity. **Figure 35.33a** shows a simplified version of the **ABC hypothesis** of flower formation, which proposes that three classes of genes direct the formation of the four types of floral organs. According to the ABC hypothesis, each class of genes is switched on in two specific whorls of the floral meristem. Normally, A genes are switched on in the two outer whorls (sepals and petals); B genes are switched on in the two middle whorls (petals and stamens); and C genes are switched on in the two inner whorls (stamens and carpels). Sepals arise from those parts of floral meristems in which only A genes are active; petals arise where A and B genes are active; stamens where B and C genes are active; and carpels where only C genes are active. The ABC hypothesis can account for the phenotypes of mutants lacking A, B, or C gene activity, with one addition: Where A gene activity is present, it inhibits C, and vice versa. If either the A gene or C gene is suppressed, the other gene is expressed. **Figure 35.33b** shows the floral patterns of mutants lacking each of the three classes of genes and depicts how the hypothesis accounts for the floral phenotypes. By constructing such hypotheses and designing experiments to test them, researchers are tracing the genetic basis of plant development.

Figure 35.33

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    The ABC hypothesis for the functioning of genes in flower
development.
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A schematic diagram of the A B C hypothesis and side views of four flowers with organ identity mutations.

Part A shows a schematic diagram of the ABC hypothesis. Three classes of organ identity genes are responsible for the spatial pattern of floral parts. These genes, designated A, B, and C, regulate expression of other genes responsible for development of sepals, petals, stamens, and carpels. Sepals develop where only A genes are expressed. Petals develop where both A and B genes are expressed. Stamens develop where both B and C genes are expressed. Carpels develop where only C genes are expressed.

Part B shows a side view of wild type flower and flowers with organ identity mutations. The phenotype of mutants lacking a functional A, B, or C organ identity gene can be explained by the model in part a and the observation that if either the A gene or C gene is suppressed, the other gene is expressed in that whorl. For example, if the A gene is suppressed in a mutant, the C gene is expressed where the A gene would normally be expressed. Therefore, carpels (C gene expressed) develop in the outermost whorl, and stamens (B and C genes expressed) develop in the next whorl.

DRAW IT*(a) For each mutant, draw a "bull's-eye" diagram like the one in part (a), labeling the type of organ and gene(s) expressed in each whorl. (b) Draw and label a "bull's-eye" diagram for a mutant flower in which the A and B genes were suppressed.*

*For suggested answer, see**Appendix A.***

In dissecting the plant to examine its parts, as we have done in this chapter, we must remember that the whole plant functions as an integrated organism. Plant structures largely reflect evolutionary adaptations to the challenges of a photoautotrophic existence on land.

Interview

Interview with Virginia Walbot: Researching plant genetics and development

A photo of Virginia Walbot.